

AYUSH GANDHI

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EDUCATION

University of Waterloo | Honors Mechatronics Engineering

Waterloo | 2025–2030

- **Awards:** President's Scholarship of Distinction.
- **Relevant Courses:** Digital Computation & Mechatronics Engineering.

Dawson College | Pure and Applied Science Program

Montréal | 2023–2025

- **Awards:** Dean's Honors List, R-score of 37
- **Relevant Courses:** Calculus I, II & III; Linear Algebra; Mechanics; Waves and Optics; and Electricity and Magnetism.

TECHNICAL SKILLS

- **Software:** C++, Python, Mediapipe, OpenCV, PyTorch, Matplotlib, NumPy, Excel
- **Hardware:** CAD (SolidWorks, Onshape, Fusion360, AutoCAD), Arduino/ESP32, ultrasonic sensors, motors & drivers.
- **Production:** Rapid Prototyping, 3D Printing, Milling Machine, Lathe, DFM, GD&T
- **Languages:** English, French, Gujarati & Hindi.

PROJECTS

6 D.O.F. Motorized Robotic Arm | Personal Project

02/2026- Present

- Designed and **3D-printed** a **6-DOF** arm with a **15 cm** reach and a **150 g** payload using **Onshape**
- Implemented **Arduino**-based joint control using **PCA9685** and **A4988**
- Developing gesture-based control with planned inverse kinematics using **OpenCV, Python & MediaPipe**

Autonomous Object Sorting Robot | University of Waterloo

10/2026-12/2026

- Boosted classification accuracy from **60% to 90%** by optimizing sensor-fused decision algorithms.
- Implemented **custom PID** control loops for precise navigation and **shortest-path routing**.
- **Modeled mechanical systems** within tight spatial constraints to improve overall hardware reliability.

Saturn-V Simulator | Dawson College

03/2025-05/2025

- Built a **Python** Saturn V simulator through multi-stage separation & fuel dynamics.
- Visualized simulation trajectories and performance metrics by analyzing historical NASA data.
- Validated outputs to authentic Saturn V flight telemetry, achieving **5%** increase through algorithms debugging

F.I.R.S.T. Robotics | École Secondaire Pierre-Laporte

09/2022–06/2023

- Designed an **800g-capacity** motorized arm in **SolidWorks**, improving mechanical efficiency by **15%** through **model analysis**.
- Managed the 20-member mechanical team, hitting 100% of build deadlines through optimized workflow planning.

WORK EXPERIENCE

Engineering and Trades Learning Assistant | Sheridan College

Brampton | 01/2026-04/2026

- Mentored **CAD**, electronics, and fabrication labs, troubleshooting designs and streamlining prototyping workflows.
- Improved design manufacturability by applying **GD&T** and **DFM** principles during CAD reviews and fabrication planning.